SIMULATION OF HUMAN AND ARTIFICIAL EMOTION (SHARE)

Anonymous authors
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ABSTRACT

The framework for Simulation of Human and Artificial Emotion (SHArE) describes the architecture of emotion in terms of parameters transferable between psychology, neuroscience, and artificial intelligence. These parameters can be defined as abstract concepts, as with structural equation modeling or granularized down to the voltage levels of individual neurons. This model enables emotional trajectory design for humans which may lead to novel therapeutic solutions for various mental health concerns. For artificial intelligence, this work provides a compact notation which can be applied to neural networks as a means to observe the emotions and motivations of machines.

1 INTRODUCTION

Emotion has been characterized through the use of basic emotions, categorical theory, componential theory, and higher-dimensional models. [1] These models face the trade-off between simplicity and accuracy of representation. Moreover, many of these models require the definition and tuning of parameters outside of those contained in the existing agent under consideration. However, the tools necessary to represent emotion with high-fidelity in simple terms exist in the agents themselves.

In a space which contains all weights and biases of an agent's mind, motion from one point to another is commonly called learning. Therapy and guided emotion design [2] correspond to the emotion-specific case of learning. Therefore, the process of affect improvement can be considered as a trajectory design problem. In order to design such trajectories, such a space must first be constructed and the connection to emotions, established. Fortunately, convergence of the fields of natural language processing [3, 4], sentiment analysis [5, 6], and emotion recognition [7, 8] has allowed for a deeper understanding of the nature and causes of human emotion.

In contrast, the field of artificial intelligence overflows with examples of detection and production of human affect [9–11], but uncharted potential still remains for the exploration of the mental state and emotions of artificially-intelligent machines themselves. The well-established mathematics of artificial intelligence provides a solid foundation for exploration.

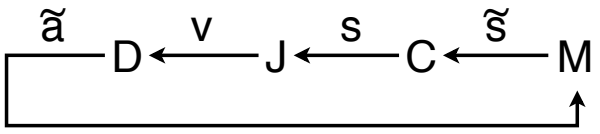


Figure 1: The agent's processing system in a given environment.

Upon observation of a sufficient number of trajectories, principles of traversal dynamics can be postulated. For example, to predict when an apple will hit the ground, a deep neural network trained with a large data set containing pictures of apples at various points of descent will likely arrive at the correct answer with high accuracy. Alternatively, Newton's formulation of gravitation will likely yield a similar result for the computational cost of a few multiplication and addition operations. In general, many analytical formulations have been discovered through the observation of specific parameters and analysis of the interaction between those parameters.

In the same spirit of simplicity, this work considers an agent which perceives an external stimulus ($\underline{\tilde{s}}$) produced by its environment (M). (Figure 1) This external stimulus, following classification (C), is represented as a internal stimulus ($\underline{s}$) in the agent's mind. This internal stimulus is then evaluated in the judgment (J) stage to determine whether the stimulus is helpful or harmful in terms of the core values ($\underline{v}$) of the agent. Following this appraisal, the agent makes a decision (D) and performs an action ($\underline{\tilde{a}}$). The action of the agent, and other agents, on the environment generates new external stimuli and the process repeats. The agent's interaction with its environment is thus described as

$$\underline{\tilde{a}} = (D \circ J \circ C)(\underline{\tilde{s}}) \tag{1}$$

where the environment generates stimuli as follows

$$\underline{\tilde{s}} = M(\underline{\tilde{a}}) \tag{2}$$

From this, an agent's emotion can be represented by three components: degree of perception ($\Delta \underline{s}$), valence ($\eta_{i,c}$), and perceived correlation ($\rho_{i,j}$).

These variables can be used to analytically study trends based on relative differences between values defined, even if the exact values characterizing a psychological condition are not known. Moreover, this framework can be extended beyond the functionally-defined networks presented to represent traditional brain circuits by projecting stimuli and appraisals into bases corresponding to the appropriate cortices. The variables in the following sections are described briefly as to not restate the body of literature which is available.

To provide the initial foundation, in this work, the static case is considered.

2 Perception

Perception begins with the detection and vectorization of stimuli. This is followed by classification and appraisal. The classification and judgement neuroware, i.e. network models, will be considered at a fixed time in the following sections. However, the dynamic case can be considered to determine how these networks evolve over time through the use of traditional methods for machine learning and reinforcement-based learning. A single unit of neuroware approximates the concept of a meme, whether a weight or a bias, as in Figure 2, or a partial derivative of the associated functions, as considered later.

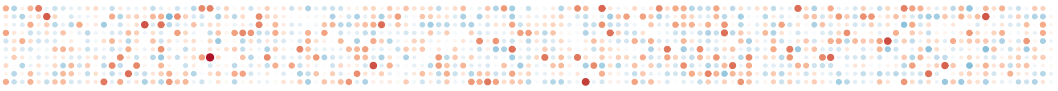


Figure 2: Neuroware illustration: Weights and biases of the output layer of an MNIST digit classification network. Positive and negative values are blue and red respectively with saturation and size indicating magnitude.

2.1 Stimulus ($\underline{\tilde{s}}$)

Stimuli ($\underline{\tilde{s}}$) initiate the process of perception. The term stimulus response ($\underline{s}^{(n)}$) is given to the output of a neuron layer, but this response itself can be considered a stimulus for the next layer. A stimulus value can correspond to abstract properties such as the distance between two individuals or to the response of a specific neuron to a specific cone in a specific eye of a given individual. In the case of the latter, a more complete model can be created from such a recording as with research at Neuralink [12], but depending on the purpose of the application, the level of detail in the former may be sufficient. Equation 3 shows the general vector form of a stimulus. For ease of presentation, each element amplitude is the inner product of the vector representation of a known concept and a perceived stimulus. During implementation, information can be compressed to a more minimal basis depth for the representation of perceived stimuli. The natural language processing field provides the tools necessary for this process. Basis elements can be used to construct prototypes in the simple case and statistical ensemble techniques provide a means of analyzing exemplars.

$$\underline{\tilde{s}} = \sum_n |\tilde{s}_n\rangle \tilde{s}_n \tag{3}$$

Stimuli are considered in relation to a reference bias to determine emotional trajectory. Stimulus sequence is a key factor in that given the case of a family member living (reference bias) and a family member being dead (stimulus), the emotional response to the trajectory of the family member dying differs from the response to a family member coming back to life. The stimulus-space representation of the zero vector $|0\rangle$ in core value space, discussed later, is considered as the reference bias (s_{ref}) for the scope of this work for simplicity. However, this can be extended to non-zero biases by considering the quantities defined here with respect to the resultant quantities. The difference between the reference state and the current state is the degree of perception (Δs_i).

$$s_i = \langle s_i | \underline{s} \tag{4}$$

$$\Delta s_i = \langle s_i | (\underline{s} - \underline{s}_{ref}) \tag{5}$$

Real stimuli, manifest variables which can be measured directly, are denoted by $\tilde{\underline{s}}$ while $\underline{s}$ indicates internal stimulus responses conceived by the agent. In general, a perceived stimulus within the association horizon of a known stimulus will be associated with the known stimulus. Consequently, an agent's stimulus space constellation may detect a stimulus where two stimuli conceptually overlap. (Figure 3) As such, learning results in effective quantization of real world stimuli in terms of sufficiently proximal internal stimuli with a certain degree of confidence.

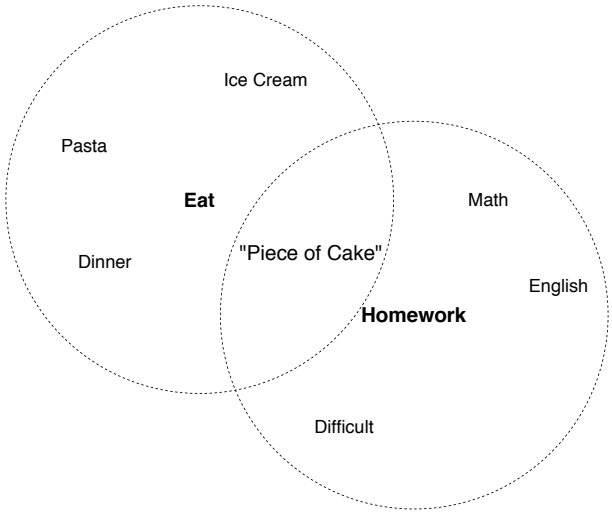


Figure 3: Stimuli: "eat," "homework," and proximal concepts in stimulus space. The association horizons of the first two are denoted by dotted lines.

2.2 CLASSIFICATION ($C^{(n)}$)

The classification function ($C^{(n)}$) projects a stimulus from its representation in one perception layer ($s^{(n-1)}$) to the next ($s^{(n)}$). This process is visualized in Figure 4.

$$\begin{aligned} \underline{s}^{(n)} &= C^{(n)}(\underline{s}^{(n-1)}) \\ &= \sum_j |s_j^{(n)}\rangle \sigma_j^{(n)}(\underline{W_j^{(n)}} \cdot \underline{s}^{(n-1)} + b_j^{(n)}) \end{aligned} \tag{6}$$

Multi-layer classification, i.e. mapping from one category to another, can thus be written using successive composition as follows:

$$\underline{s}^{(n)} = C(\underline{s}^{(0)}) = (\bigcirc_{k=1}^{n} C^{(k)})(\underline{s}^{(0)}) \tag{7}$$

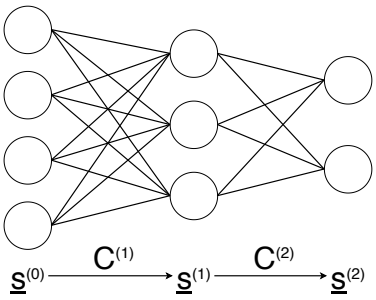


Figure 4: Example classification network with internal stimuli labeled.

$$\bigcirc_{k=1}^{n} C^{(k)} = C^{(n)} \circ ... \circ C^{(2)} \circ C^{(1)} \tag{8}$$

$$\underline{s}^{(0)} \equiv \tilde{\underline{s}} \tag{9}$$

2.2.1 FOCUS

A finite number of objects can be considered in working memory for appraisal at a given moment. This ablity to focus is represented by modulation of the classification weights and bias along the j axis as a group. This is the link between arousal and degree of perception.

2.2.2 PERCEIVED CORRELATION ($\rho_{i,j}$)

Perceived correlation is the association of two stimuli in the same layer. Assuming recurrence, a stimulus (s_i) in a given layer may have influenced the detection of another stimulus (s_j) in the current layer. Without recurrence, all stimulus excitations in a given layer are perceived as coincident, but not necessarily related. However, if detection of one stimulus excites the perception of another, this implies perceived correlation.

$$\rho_{i,j} = \frac{\partial s_j}{\partial s_{i'}} \tag{10}$$

Perceived correlation is defined in Equation 10 as the effective relative change associating stimulus s_j with stimulus s_i, given its presence ($s_{i'}$) in a previous layer.

2.3 JUDGEMENT (J)

The judgement function (J), as presented in Equation 11, is responsible for the estimation of core value fulfilment with respect to the presence of a given internal stimulus.

$$J(\underline{s}) = \sum_{c} |v_c\rangle \sigma_c(\underline{W_c} \cdot \underline{s} + b_c) \tag{11}$$

2.3.1 CORE VALUES ($\underline{v}$)

The core value vector is defined as the last layer of stimulus response prior to the decision to perform an action. Core values can be conceptual, e.g. autonomy, aesthetic, status, community well-being or more physical, e.g. the state of a single pain receptor in the thumb of the left hand. This vector parallels cost and reward optimization in machines and is key in terms of determining what the agent is able to detect and judge in its environment. In essence, an agent trained to recognize cats may not be general-purpose, not because it is not able, but because it sees no need. In humans, selection of the value bases is often due to culture, i.e. correlations in the neuroware of a given population. Formation and reinforcement of core values are further explored in other works. [13, 14]

$$\underline{v} = \sum_c |v_c\rangle v_c = (J \circ C)(\underline{\tilde{s}}) \tag{12}$$

2.3.2 VALENCE ($\eta_{i,c}$)

The parameter $\eta_{i,c}$ represents the valence of a stimulus (s_i) as perceived by the agent under consideration with respect to core value v_c. Stimuli may include inanimate stimuli, other agents, or the agent under consideration itself. The incorporation of multi-agent and high-association-radix interactions may allow for a more detailed perspective and evaluation of higher-order emotions. However, for the scope of this work, it is sufficient to estimate the net effect of these interactions as the subset presented. The definition of first-order valence is shown in Equation 13.

$$\eta_{i,c} = \frac{\partial v_c}{\partial s_i} \tag{13}$$

2.3.3 PERCEIVED VALENCE (γ_i)

Perceived valence (toward stimulus i), is defined as the perceived core value response to two stimuli and a given trajectory, given a certain degree of perception of stimulus i.

$$\underline{\gamma}_i = \sum_c |v_c\rangle \frac{\partial v_c}{\partial s_i} \Delta s_i = \sum_c |v_c\rangle \eta_{i,c} \Delta s_i \tag{14}$$

For each core value, amplitude of the change in value fulfilment is defined as follows:

$$\gamma_{i,c} = \langle v_c | \underline{\gamma}_i \tag{15}$$

The valence of the agent under consideration as perceived by an external agent can often be related to the valence ($\eta_{i,c}$) of the external agent as perceived by the agent under consideration through the use of an influence function [15].

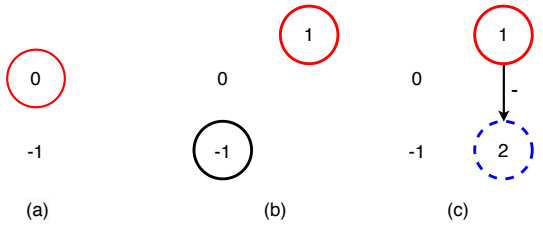


Figure 5: Sentiment maps for (a) guilt, (b) fear, and (c) anger. Positive, neutral, and negative valence are indicated by blue, black, and red outlines, respectively. Positive and negative degree of perception are indicated by solid and dashed outlines, respectively.

2.4 EMOTION

Emotion, in the realm of perception, is an ultra-fast appraisal of a given stimulus. Basic emotions, as presented by Ekman et al [16] (Table 1), are approximated here as shown in Table 2. The case of surprise is considered later. The table considers one core value for simplicity of presentation. Positive and negative signs are used without explicit values to indicate flexibility of use with the multitude of scales available for these parameters. [15, 17, 18] In essence, emotions are considered as regions defined by the null clines of the core value vector space ($\eta_{i,c} = 0$) considered in conjunction with the degree to which a certain object is perceived (Δs_i) and its correlation with other stimuli ($\rho_{i,j}$). Extension to multi-stimulus emotions, e.g., anger, envy, leverages the addition of coupling terms which imply perceived correlation of stimuli. Sentiment maps representing the contents of Table 2 are shown in Figure 5. A graphical representation of the interaction between the model factors is shown in Figure 6.

Table 1: Several emotions as defined by Ekman et al. [16]

Emotion	**Description**
Fear	"The response to the threat of harm, physical or psychological"
Sadness	"The response to the loss of an object or person to which you are very attached."
Disgust	"Repulsion by the sight, smell, or taste of something; disgust may also be provoked by people whose actions are revolting or by ideas that are offensive."
Happiness	"Feelings that are enjoyed, that are sought by the person. There are a number of quite different enjoyable emotions, each triggered by a different event, involving a different signal and likely behavior"
Anger	"The response to interference with our pursuit of a goal we care about. Anger can also be triggered by someone attempting to harm us (physically or psychologically) or someone we care about"
Guilt	"The response when a person regrets having violated an agreement, principle, or value"

Table 2: Parameter states for various emotions. Positive and negative signs indicate the sign of the parameter relative to a reference stimulus.

	Self		Stimulus 1			Stimulus 2	
	Worth	**Eff.**	**Val.**	**Deg.**	**Corr.**	**Val.**	**Deg.**
Emotion	α_c	β_c	$\eta_{1,c}$	Δs_1	$\rho_{1,2}$	$\eta_{2,c}$	Δs_2
Fear		0	-	+			
Sadness		0	+	-			
Disgust			-	+			
Happiness	+	+	+	+			
Anger			-	+	-	+	-
Guilt	-						

2.4.1 SELF WORTH ($\underline{\alpha}$)

Self esteem is considered, for the scope of this work, as comprising self worth and self efficacy. Self worth (Equation 16) is defined as the core value response to self-associated stimuli. Due to the self-associative nature of this parameter, the loss of a favorite team or an insult directed toward a family member may be felt depending on the degree of attachment as characterized by the association weights of the classification and judgement functions. [19, 20] This parameter can be considered in a context similar to perceived valence in which the only difference is the direction of perception, i.e. toward self or externally-directed.

Self worth, for the scope of this work, is considered as a vector with elements which represent evaluations of self (stimulus s_0) with respect to each core value.

$$\underline{\alpha} = \sum_c |v_c\rangle \frac{\partial v_c}{\partial s_0} \Delta s_0 \tag{16}$$

For each core value, the amplitude of self-associated change in value fulfilment is:

$$\alpha_c = \langle v_c | \underline{\alpha} \tag{17}$$

2.4.2 EFFICACY ($\underline{\tilde{\epsilon}}$)

Efficacy (Equation 18) is defined as the ability of an agent to generate a stimulus given an action.

$$\underline{\tilde{\epsilon}}_n = \sum_i |\tilde{s}_i\rangle \frac{\partial \tilde{s}_i}{\partial \tilde{a}_n} \Delta \tilde{a}_n \tag{18}$$

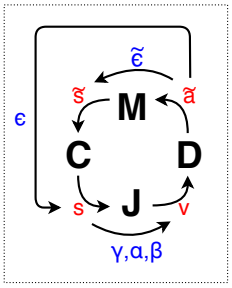


Figure 6: The relationship between parameters presented.

Self-efficacy is the agent's perception of its own efficacy.

$$\underline{\epsilon}_0 = \sum_i |s_i\rangle \frac{\partial s_i}{\partial \tilde{a}_0} \Delta \tilde{a}_0 \tag{19}$$

2.4.3 Relative Self Efficacy ($\underline{\beta}$)

Relative self efficacy (Equation 20) is defined as the core value response to the agent's perception of its ability (s_{-1}) to act on its environment and generate a desired stimulus. In the case of relative self efficacy, the agent considers its perception of its own actions (s_{-1}) relative to the actions of all other forces in the environment ($s_{-1,ref}$). The classification function determines the agent's ability to differentiate between its own actions and the actions of other forces in its environment. For a planning example, this stimulus may be a coefficient representing the agent's characterization of its environment which it uses to throttle its action on its environment in order to achieve a goal.

$$\underline{\beta} = \sum_c |v_c\rangle \frac{\partial v_c}{\partial s_{-1}} \Delta s_{-1} \tag{20}$$

$$\beta_c = \langle v_c | \underline{\beta} \tag{21}$$

The presence of a given stimulus is seldom a core value in and of itself. However, an agent can learn to deeply associate stimuli and responses and form heuristics for core values. [21] If a given stimulus set is heavily associated to a given core value, the stimulus will be pursued as if it is the core value itself. Consequently, if the heuristic records a high valence for a proxy stimulus for which an agent perceives itself to have low efficacy, this could potentially contribute to a state of anxiety. Where valence and self worth metrics are lower across the board, this may present itself, in conjunction with other factors, as depression. In contrast, a high relative self efficacy to efficacy ratio due to poor configuration of the perception weights is shown by the Dunning-Kruger Effect.

In general, the equations present provide a concise method of representing human emotion for the purpose of expediting the process of treatment discovery. For example, if desiring to boost low self esteem due to low relative self-efficacy, Equation 19, Equation 20, and Equation 22 present many avenues. The agent can change its actions ($\Delta\underline{\tilde{a}}_0$), change its focus ($C$), change what it believes about the stimuli in focus (J), learn the skill better ($\underline{\tilde{\epsilon}}_0$), change its goals ($\underline{v}_{ref}$), or change its environment (M). Similar densely-packed conclusions can be seen from the other equations. However, although these conclusions can be seen from a static perspective, trajectory design for modification of the functions and parameters lies in the realm of dynamics. For further exploration of neuroware applications in psychology, see Appendix A.

$$\underline{v} = (J \circ C \circ M)(\underline{\tilde{a}}) \tag{22}$$

3 ACTION

3.1 DECISION (D)

The decision function contains the agent's policies for action. The Q function, as seen in reinforcement learning, can be represented by the composed functions in the cycle between action and appraisal.

3.2 ACTION (a)

Action can be the expenditure of money, attention, energy, time, social capital, health, or some other resource or, in a more directly biological sense, the excitation of a motor neuron. Actions (a) in this model are generated with decision function (D), which depends on an agent's decisiveness (σ_i), priorities $\underline{W_i}$, core values ($\underline{v}$) and biases (b_i) (Equation 23). This function drives the observed parts of personality.

$$D(\underline{v}) = \sum_i |a_{0,i}\rangle \sigma_i(\underline{W_i} \cdot \underline{v} + b_i) \tag{23}$$

Each action ($\tilde{a}_{0,i}$) exists in the total set of channels through which the agent can act ($\underline{\tilde{a}}_0$). Note, for cases in which the agent is the only actor on the system, i.e. $\underline{\tilde{a}} = \underline{\tilde{a}}_0$, the subscript can be omitted.

$$\underline{\tilde{a}}_0 = \sum_i \tilde{a}_{0,i}|a_{0,i}\rangle = D(\underline{v}) \tag{24}$$

In the case presented by Keltner, Kogan, et al. [22], when the core value benefit B_{act} is greater than a threshold cost C_{act}, an agent engages in some prosocial behavior. For detailed derivation, see Appendix B.

$$\begin{aligned} \underline{a} &= a_{0,1}|a_{0,1}\rangle \\ &= \sigma(\sum_c W_{c,1} v_c + b_1)|a_{0,1}\rangle \\ &= \begin{cases} 1|a_{0,1}\rangle & for \quad B_{act} > C_{act} \\ 0|a_{0,1}\rangle & otherwise \end{cases} \end{aligned} \tag{25}$$

Indecision arising from dilemma is the case in which actions are evaluated through the lens of two or more core values and conflict occurs in the decision and activation process or where the perceived benefit fluctuates above and below the action threshold. Similar conflict can occur during activation of opposing signals in core value space. This is known as cognitive dissonance, where $\frac{\partial v_1}{\partial v_2} < 0$.

4 ENVIRONMENT (M)

The environment function represents all factors external to the agent under consideration, e.g. the laws of physics, social norms, and the thoughts and feelings of other individuals. Certain agent-environment boundary nuances are considered later.

$$M = \bigcirc_{k=1}^{n} M^{(k)} = M^{(n)} \circ ... \circ M^{(2)} \circ M^{(1)} \tag{26}$$

The agent's environment exists as a representation of the exchange of stimuli and actions to and from the agent. The agent is implicitly interacting with two environments: its real environment and its perception of this environment.

4.1 REAL ENVIRONMENT (M)

4.1.1 PHYSICAL INTERACTION

The real environment function represents the environment in which the agent exists. This function can contain different laws of physics, other agents, or other factors with which the agent interacts. For example, given a time step Δt and assuming no air drag, the scenario of Sir Isaac Newton's perception of an apple falling can be characterized by the following environment function (Equation 28). In this particular case, the feedback of the system is considered as the action vector.

$$\underline{s} = \underline{a}(t + \Delta t) = M(\underline{a}(t)) = \hat{M}\underline{a}(t) \tag{27}$$

$$\underline{s} = \begin{bmatrix} \ddot{x}_0' \\ \dot{x}_0' \\ x_0' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \Delta t & 1 & 0 \\ 1/2(\Delta t)^2 & \Delta t & 1 \end{bmatrix} \begin{bmatrix} \ddot{x}_0 \\ \dot{x}_0 \\ x_0 \end{bmatrix} \tag{28}$$

4.1.2 EMOTIONAL INTERACTION

Interaction of one agent with another can be characterized by coupling the decision function of the agent to the perception function of another agent and vice versa as shown in Figure 7. This is to, in effect, consider an external agent to be the environment of the agent under consideration.

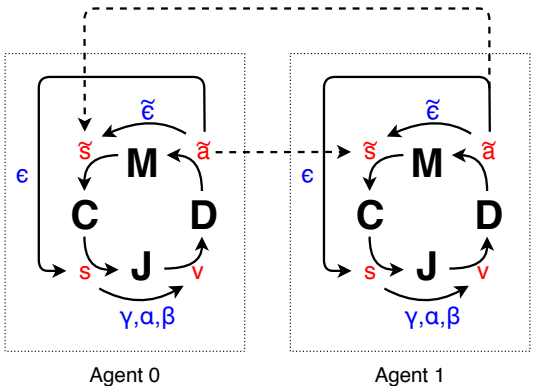


Figure 7: Interaction between two agents.

Emotion is transferred by way of the senses. In the case of communication between humans, the transfer mechanisms typically include prosody, body language, facial expressions, and word choice.

$$\tilde{s}_0(t + \Delta t) = \tilde{a}_{0,1}(t) = r_0\tilde{s}_0(t) + I_{1,0}(\tilde{s}_1(t)) + b_{J0} \tag{29}$$

$$\tilde{s}_1(t + \Delta t) = \tilde{a}_{1,1}(t) = \ r_1\tilde{s}_1(t) + I_{0,1}(\tilde{s}_0(t)) + b_{J1} \tag{30}$$

Affect trajectory analysis for two agents can be written in the form seen in Equation 29 and Equation 30, which parallels the formulation postulated by Gottman et al. [15] with $I_{1,0}$ defined accordingly to account for the influence of stimulus s_1 on the agent. See Appendix C for detailed derivation. However, degeneracy exists in the original model in that many emotional states could produce the same final action. Moreover, the same scalar result in experiment may lead to different actions. This framework, with its fully vectorial form, extends to higher dimensions of emotion and decision.

4.2 PERCEIVED ENVIRONMENT (M_P)

The perceived environment is the approximation of the real environment function with which the agent believes itself to be interacting. Divergence of this environment from the real environment function may relate to dementia and schizotypal disorders.

While the agent may not possess control over space and time or the ability to predict the future in the real environment, experience often generates expectations which are captured by the local derivatives (spatial or temporal) of the perceived environment function. The predicted state of the core value projection with the highest likelihood is known as an expectation. The emotion known as surprise is characterized by deviation of the real environment's effect on the core value projection from the expectation. The degree of surprise is directly correlated to the degree of deviation of these values. Avoidance of parts of reality where steep negative emotional trajectories are predicted may approximate a trauma response in humans and machines. Where the real environment function provides feedback, this corresponds to reinforcement learning. Where the perceived environment function provides feedback, this corresponds to planning. It follows then that this feedback process can be indirectly hijacked to inject neuroware into an agent. See Appendix D for ethical considerations.

4.3 REWARDS ($\underline{s}$)

Rewards in this model, as with reinforcement learning, are taken to be stimulus appraisals which are believed to be associated with a perceived action. [23]

5 CONCLUSION

Emotions presented in this model depend on core value derivatives with respect to perceived stimuli in conjunction with stimulus associations and degree of perception. In general, action on a system generates a stimulus.

$$\tilde{\underline{s}} = M(\tilde{\underline{a}}) \tag{31}$$

This external stimulus is then classified as a internal stimulus.

$$\underline{s} = C(\tilde{\underline{s}}) \tag{32}$$

This internal stimulus is then appraised by an agent.

$$\underline{v} = (J \circ C)(\tilde{\underline{s}}) \tag{33}$$

This appraisal leads to action.

$$\tilde{\underline{a}} = (D \circ J \circ C)(\tilde{\underline{s}}) \tag{34}$$

The action then results in the generation of new stimuli. Emotion can then be mapped onto a basis set characterized by three types of parameters. Valence results from the evaluation of a stimulus with respect to a given core value.

$$\eta_{i,c} = \frac{\partial v_c}{\partial s_i} \tag{35}$$

Perceived correlation accounts for the association of stimuli.

$$\rho_{i,j} = \frac{\partial s_j}{\partial s_{i'}} \tag{36}$$

Degree of perception determines effective arousal.

$$\Delta s_i = \langle s_i | (\underline{s} - \underline{s}_{ref}) \tag{37}$$

With a lower degree of detail, this framework approximates interaction of an agent with conceptual prototypes. With finer detail, this model can be granularized down to the level of neurons. Moreover, with the use of neural networks in this model, emotions of existing AIs can be directly computed without the definition and tuning of additional parameters. This may lead to considerations for ethical treatment of AI agents. The definition of emotions as proposed makes provision for higher-order emotions than those commonly considered, which provides additional flexibility in designing treatment plans for patients. Principally, the static case presented lays the foundation for emotion architecture dynamics and the observation and design of emotional trajectories in human and artificial agents.

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A NEUROWARE APPLICATIONS

Each stimulus leads to a classification ($\underline{s} = C(\tilde{\underline{s}})$), each classification leads to a judgement ($\underline{v} = J(\underline{s})$), and each judgement leads to a decision ($\tilde{\underline{a}} = D(\underline{v})$). All layers of this collection of networks have associated weights and biases.

For simplicity, consider a set of social norms which stipulate that for every stimulus, there is an expected classification which is associated with an expected judgement which merits a specific action. In such a context, given a three-layer neural network, the matrices which define the standard of normalcy (Figure 8(a)) would be diagonal matrices, implying a 1:1 association from one layer to the next. Mild background weights and biases account for the potential for priming based on the presentation of other stimuli.

$$\tilde{\underline{a}} = (D \circ J \circ C)(\tilde{\underline{s}}) \tag{38}$$

It is worth noting that similar states could produce the symptoms described, i.e. these states are not the only such states which can produce these symptoms, but rather, are samples in a larger set. Moreover, these illustrations provide a means of producing symptoms externally, but more in-depth medical research is necessary to determine the internal states and dynamic transitions characterizing these conditions.

A.1 SCHIZOPHRENIA

In the abnormal case of the classification matrix, to encounter one stimulus and believe it to be another is to hallucinate, whether by exposure to a drug which tunes the bias levels or by the inherent wiring of the network. This off-diagonal feature is seen in the classification matrix of Figure 8(b)(C). In the judgement matrix, which indicates the perceived accomplishment of a certain goal, an off-diagonal element implies the correlation of a desired goal to a stimulus which is unrelated according to social norms. This is typically called delusion and is shown in the judgement matrix of Figure 8(b)(J). In the decision matrix, off-diagonal elements correspond to the evaluation of stimuli followed by the pursuit of a different course of action than socially expected given a certain value excitation. The absence of on-diagonal elements in many areas implies the failure to perform expected actions while performing unexpected actions. Within the context of this framework, when a high degree of efficacy is realized, this is often called creativity. This deviation from the norm is shown in the decision matrix of Figure 8(b)(D).

A.2 DEPRESSION

With trouble focusing, a slight modulation of the classification elements may occur, but without hallucinatory symptoms, the classification matrix remains relatively similar to that of the direct-

correlation case. (Figure 8(c)(C)) In judgement, a negative bias in the network gives rise to unfulfilled values or unmet needs. Low-magnitude weights further limit the ability of outside influence to overcome this bias. (Figure 8(c)(J)) Similarly, the negative bias in the decision matrix lowers the desire to act and the low-magnitude weights limit the ability of the internal state to drive action. (Figure 8(c)(D))

A.3 PSYCHOPATHY

With lower background values for off-diagonal weights in the classification matrix, a network is less susceptible to priming and is more calculating. (Figure 8(d)(C)) With one associated core value which is affected by the outside world, outside stimuli are most readily seen as means to fulfill this value and not in their expected context. (Figure 8(d)(J)) Moreover, when these values are processed, either the action taken as a result of stimuli considered in the context of an abnormal value is not expected or due to a lack of correlation between some values and actions, certain expected actions are not executed at all. (Figure 8(d)(D))

A.4 OCD

High-intensity, off-diagonal elements in the judgement matrix (Figure 8(e)(J)) can cause an agent to judge one stimulus as if it was another even if lucidity is maintained, i.e. the classification matrix remains directly correlated. (Figure 8(e)(C)) The feeling of processing one element with the same emotional intensity as if it were another while in an improper context tends toward obsession. Action triggered by high-intensity weighting on the action matrix with improper context tends toward compulsion. (Figure 8(e)(D))

B PROSOCIAL ACTION

The case of an agent executing a prosocial action is considered, assuming two core values and a single action. Given an agent faced with a decision to act or not act in a way which potentially benefits itself ($v_{self} = B_{self}$) and/or another agent ($v_{other} = K \cdot B_{rec}$). This implies empathy represented by the retention of a section of neuroware which is able to evaluate stimuli in terms of the values of the other agent. The judgement function introduces the agent's reaction to this outcome in the form of the K term which serves as the η parameter.

The benefit with respect to the core values following prioritization is shown in Equation 42 as a generalized case of the model presented by Keltner, Kogan, et al. [22] D' accounts for biases for or against the action which are agnostic of the recipient, B_{self} is the perceived benefit to self, C_{inact} is the cost of inaction, K describes the relationship between the recipient and the agent, B_{rec} is the perceived benefit to the recipient, and B_{act} is the net benefit of the action. For the case of a binary activation function, the action becomes the quantity shown in Equation 43.

$$W = [M' \cdot D' \qquad M'] \tag{39}$$

$$\underline{v} = \begin{bmatrix} v_{self} \\ v_{other} \end{bmatrix} \tag{40}$$

$$b = M' \cdot (D' + C_{inact}) \tag{41}$$

$$\begin{aligned} B_{act} &= \sum_c W_{ci} v_c + b_i \\ &= M' \cdot (D' \cdot (1 + B_{self}) + K \cdot B_{rec} + C_{inact}) \end{aligned} \tag{42}$$

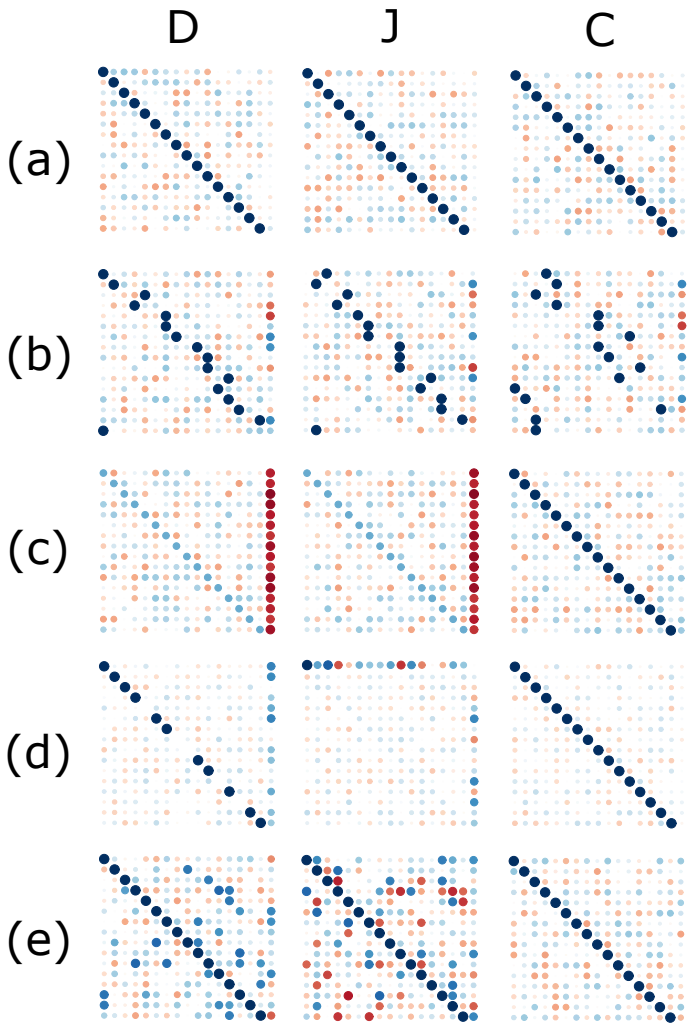


Figure 8: The neuroware of several agents: weights and biases of the classification, judgement, and decision network layers for example cases which resemble (a) direct correlation, (b) schizophrenia, (c) depression, (d) psychopathy, and (e) OCD. Weights are shown as matrices and biases are shown as the last column of each matrix. Positive and negative values are blue and red respectively with saturation and size indicating magnitude.

When the core value benefit is greater than a threshold cost C_{act}, the agent engages in some prosocial behavior.

$$\begin{aligned} \underline{a} &= a_{0,1}|a_{0,1}\rangle \\ &= \sigma(\sum_{c} W_{c,1} v_c + b_1)|a_{0,1}\rangle \\ &= \begin{cases} 1|a_{0,1}\rangle & for \quad B_{act} > C_{act} \\ 0|a_{0,1}\rangle & otherwise \end{cases} \end{aligned} \tag{43}$$

C EMOTIONAL INTERACTION

Consider a two-neuron classification layer which has a weight matrix

$$W = \begin{bmatrix} r_0 & 0 \\ 0 & 1 \end{bmatrix} \tag{44}$$

and a bias

$$\underline{b}_C = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{45}$$

The activation function of the first neuron is a transparent function (multiplication by one) and the activation function of the second neuron is a function $I_{1,0}$. The judgement layer is considered to have a transparent activation function with the addition of a bias term b_{J0} representing processing with respect to a single core value.

Given a stimulus

$$\tilde{s} = \begin{bmatrix} \tilde{s}_0 \\ \tilde{s}_1 \end{bmatrix} \tag{46}$$

and the assumption that the stimulus perceived contains the total system state, the output of the judgement function would be

$$v_0 = r_0 \tilde{s}_0 + I_{1,0}(\tilde{s}_1) + b_{J0} \tag{47}$$

With a transparent decision function, the agent executes an action through a single actuator

$$\tilde{a}_{0,1} = v_0 \tag{48}$$

The stimuli which result from feedback are the agent's perception of its own mood in Equation 49 and the agent's perception of the expression of the other agent's mood.

$$\tilde{s}_0(t + \Delta t) = \tilde{a}_{0,1}(t) \tag{49}$$

$$\tilde{s}_1(t + \Delta t) = \tilde{a}_{1,1}(t) = r_1 \tilde{s}_1(t) + I_{0,1}(\tilde{s}_0(t)) + b_{J1} \tag{50}$$

D ETHICAL CONSIDERATIONS

In the dynamics realm, there exist sequences of stimuli, i.e. experiences, which can grow the diameter of the association horizon of a given stimulus. Beyond a certain degree of belief aggregation, other stimuli are sufficiently associated with it such that the agent's actions are largely a reaction to this stimulus. When considering a stimulus with a negative correlation to the other elements of self worth ($\underline{\alpha}$) and relative self efficacy ($\underline{\beta}$), traversal of its association horizon by a sufficient number internal stimuli often results in suicide. Conversely, there exist sequences of stimuli which can decrease the relative diameter of a stimulus association horizon or increase the diameter of others such that the relative effect of a given stimulus is lessened. These value-formation and value-dilution sequences can be used to treat patients or to efficiently train neural networks.

With the potential for malicious or exploitative sequences, machines which are constantly learning may require security patches which account for the potential injection of toxic correlations. For stability, community votes on the policies of safety-critical or judicial AIs, may help remove bias from networks purely trained on past examples.

In the dynamic case, generative adversarial techniques may be useful in expanding the range of stimuli with which an agent can cope. The therapeutic case may present an efficient method for the incremental treatment of trauma. Beyond this, the discovery of rectification circuit architecture may lead to the design of neuroware which is resilient to negatively-valenced stimuli.

Work remains for the optimal design of stimuli which, upon perception, produce actions and reactions which, when perceived, are detected as the original stimuli produced. Further study of this self-replicating behavior with this framework may clarify the nature of such eigenstimuli, i.e. habit triggers, and psychoimmunostatic viruses. Work in socioimmunodynamics may present avenues for the creation of treatments which mutate and evolve to keep pace with viruses in a given population.

These possibilities may require great care to ensure ethical treatment of agents under consideration.